

LUKE BUONANNO

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PROFESSIONAL PROFILE

Mechanical Engineering student concentrating in Computer Science and Mathematics, combining hands-on CAD and product-design experience with programming, embedded systems, and hardware troubleshooting skills. Builds independent hardware and firmware projects and applies strong logic-planning and debugging skills across languages. Proven responsible leader experienced in managing communities, mediating conflict, and coordinating programs.

EDUCATION

King's College, Wilkes-Barre, PA

Anticipated Graduation: 2028

Bachelor of Science in Mechanical Engineering | GPA: 3.5

Concentrations in Computer Science and Mathematics

Dean's List: Fall 2024, Spring 2025

TECHNICAL SKILLS

CAD & Design: Autodesk Inventor (advanced, 4+ years, 3D design & drafting), AutoCAD, SolidWorks (working knowledge), Cura slicer

Programming: HTML5 (advanced), Discord.js / Discord.py / Node.js (advanced), Python, C++, Java (intermediate); strong comprehensive project development, debugging, and logic/systems planning across languages

Hardware & Lab: LabVIEW, Arduino IDE, multimeter/oscilloscope use, schematic reading, breadboarding, soldering

Tools & Platforms: Git, Linux, Windows, Microsoft Office, VS Code, Eclipse

Languages: Intermediate Spanish

PROJECTS

Lemony Studios — Founder & Designer

Ongoing

- Designed complete embedded systems around ESP32 microcontrollers integrating LiPo power management, displays, user input, and custom mechanical enclosures. Produced CAD models and manufactured functional prototypes using additive manufacturing. Developed firmware integrating all subsystems.
- Designing an embodied AI robot on a Raspberry Pi 5 paired with a dedicated desktop AI server networked for real-time intelligent movement, decision-making, and interaction
- Architecting a modular “Brain” system for long-term memory and context management; built out the mic, input, display, LLM, Tailscale connection, and speech-to-text pipeline already
- Designed and maintain technical portfolio website (lemonystudios.onthewifi.com) documenting embedded systems, CAD designs, firmware development, and engineering projects using HTML/CSS.
- Conceptually advised a University of Pittsburgh research colleague on a diamond anvil cell research setup, proposing a simplified sample-access solution to replace an existing 16-bolt assembly

HenHacks Hackathon, University of Delaware

- Collaborated on teams to design, build, and present two projects, “Dose Doodle” (a prescription medicine assistant) and “Hen-tersection,” (the intersection reimagined for pedestrians) each within a 24-hour hackathon timeframe (<https://devpost.com/LukeBuonanno>)

PROFESSIONAL EXPERIENCE

King's College Resident Life, Wilkes-Barre, PA

August 2022 – Present

Resident Assistant

- Provide structure and support to a community of 13–25 residents, many attending college for the first time, mediating peer conflicts and enforcing Residence Life policies
- Plan and lead community programs and events that foster connection and support among residents

IITS, King's College, Wilkes-Barre, PA

October 2024 – March 2025

IT Student Support Specialist

- Resolved hardware, software, networking and printer-related issues for students and faculty while maintaining rapid response times across campus support requests.

Wegmans Food Markets, Allentown, PA

August 2022 – Present

Front End Employee

- Processed payments and operated the cash register 20% faster than the store average
- Provided product knowledge, resolved complaints, and responded to inquiries to increase customer satisfaction
- Received 50+ customer commendations to management for service quality

St. Luke's Hospital

May 2022 – May 2024

Volunteer Patient Advocate

- Supported patients by offering companionship and providing comfort items such as drinks and blankets upon request

ACTIVITIES, LEADERSHIP & AWARDS

- Treasurer, Engineering Club, King's College | Vice President, Coding Club, King's College